HAO ZHANG

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Research Interests: Expressive Speech Translation, Full-Duplex Spoken Dialogue & Conversational Naturalness, Low-Resource Speech

EDUCATION

Johns Hopkins University, USA, MSE in Computer Science 01/2025-12/2026
- GPA: **3.88**/4; Human Language Technology Concentration (Thesis Based)
- Courses: Machine Learning for Signal Processing (A+), Information Retrieval and Web Agents (A), Introduction to Human Language Technology (A+), Natural Language Processing (A), Information Extraction (A+)

Shandong University, China, B.Eng in Intelligent Medical Engineering 09/2020-07/2024
- GPA: **3.74**/4
- Courses: Python Programming and Introduction to Algorithms (A), Signal Analysis and Processing (A), Digital Image Processing (A), Healthcare Big Data Analytics (A), Cognitive Science and Brain-like Computing (A), Machine Learning (A)
- Outstanding Graduate, Shandong University (Top 5 in Major) 2024, Academic Scholarship in 2022, 2023

PAPER & PATENT

Full-Duplex Dialogue & Conversational Naturalness (M.S.E.) (†indicates equal contribution; ‡indicates corresponding author)

[1] **Hao Zhang**, ..., Laureano Moro-Velázquez, CONTACT: A Human-Grounded Benchmark and Surprisal-Based Predictive Scorer for Conversational Naturalness. *The Fifteenth International Conference on Learning Representations (ICLR)*, 2026.9 (**Under Review**)

[2] **Hao Zhang**†,‡, Sathvik Napa Ugandhar†,‡, ..., Laureano Moro-Velázquez‡, TRACE: Temporal Relationship-Aware Conversational Entrainment Detection in Dyadic Speech. *IEEE Spoken Language Technology Workshop (SLT)*, 2026.8 (**Accepted**), arXiv:2606.30543

[3] **Hao Zhang**, Thomas Thebaud, Georgi Tinchev, Venkatesh Ravichandran, Laureano Moro-Velázquez, TurnNat: Automatic Evaluation of Turn-Taking Naturalness in Dyadic Spoken Dialogue. *arXiv preprint*, 2026.7, arXiv:2607.01345, Demo and Code

[4] Thomas Thebaud, Yuzhe Wang, **Hao Zhang**, ..., Laureano Moro-Velázquez, SPEARBench: A Benchmark for Naturalness Evaluation in Streaming Speech-to-Speech Language Models. *Computer Speech & Language (CSL)*, 2026.7 (**Under Review**), arXiv:2607.05365

Expressive Speech Generation & Translation (M.S.E.)

[5] **Hao Zhang**, ..., Berrak Sisman, Philipp Koehn, Holistic Parallel Supervision for Expressive Speech to Speech Translation. *2027 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.9 (**Under Review**), Demo and Code

[6] Zihan Zhang, ..., **Hao Zhang**, Philipp Koehn, Berrak Sisman, TRACE-EVC: Text-Guided Relative Affective Control for Zero-Shot Emotional Voice Conversion. *IEEE Open Journal of Signal Processing (OJSP)*, 2026.9 (**Under Review**), arXiv:2607.03666

Other Work: Sentiment Analysis (M.S.E.); Time Series & Signal Processing (B.Eng.)

[7] **Hao Zhang**, Rui Liu, Chengzhe Li, Philipp Koehn, Learning Sentiment Dynamics: Knowledge-Augmented Sequence Modeling for Document-Level Aspect-Based Sentiment Analysis. *The 2027 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2026.10 (**Under Review**)

[8] **Hao Zhang**, Teng Li‡, Ai Jing, Siyuan Yang, Sequence-Spectrogram Fusion Network for Wind Turbine Diagnosis through Few-Shot Time Series Classification. *Advanced Engineering Informatics (AEI)*, (Impact Factor: 11.5), 2025.1 (**Published**), Code and Datasets

[9] **Hao Zhang**, Zhendong Pang, Jiangpeng Wang, Teng Li‡, Few-Shot Learning Using Data Augmentation and Time-Frequency Transformation for Time Series Classification. *5th International Conference on Robotics, Intelligent Control and Artificial Intelligence (RICAI)*, IEEE, 2023.12 (**Published, Oral**)

[10] Zhendong Pang, **Hao Zhang**, Teng Li‡, Hybrid Fine-Tuning in Large Language Model Learning for Machinery Fault Diagnosis. *22nd IEEE International Conference on Industrial Informatics (INDIN)*, 2024.6 (**Published**)

[11] Bingyu Yang, ..., **Hao Zhang**, Teng Li‡, Hierarchical Informative Path Planning for Active Sensing and Field Reconstruction. *Proceedings of the 2024 3rd International Symposium on Control Engineering and Robotics (ISCER)*, ACM, 2024.8 (**Published**)

Invention Patent A Method and System for Wind Turbine Condition Monitoring Based on Multimodal Feature Fusion. Chinese Invention Patent CN118211123B, 2024.8 (**Granted**)

ACADEMIC RESEARCH

***Research Assistant,* Center for Language and Speech Processing (CLSP)** 11/2025 - Present
Supervisor: Prof. Laureano Moro-Velázquez, Johns Hopkins University (projects funded by Amazon)

❖ *[ICLR 2027, under review]* **CONTACT: Human-Grounded Benchmark & Scorer for Conversational Naturalness** 11/2025-09/2026
- Existing full-duplex benchmarks score turn-taking and affect separately and lack real conversations rated as unnatural; therefore proposed CONTACT to test whether automatic metrics match human perception.
- Led an IRB-approved collection of **493** rated conversations (**39.8 h**), from elicited human–human disruptions to live sessions with six voice agents, designing the protocol, rubrics and quality checks, and auditing sessions.
- Extended TurnNat into a surprisal-based scorer with future joint-silence and arousal–valence targets to capture withheld responses and affect mismatch, beating seven baselines in agreement with human ratings (overall $\rho = \mathbf{0.36}$ vs. 0.26 for the best).

❖ *[SLT 2026, co-first author]* **TRACE: Temporal Relationship-Aware Conversational Entrainment Detection** 12/2025-06/2026
- Emotional entrainment depends on relationship and context, yet no data separate entrained from non-entrained dyads; therefore co-proposed DyadEE, the first such dataset (8,434 dyads, 1,070 h), built by emotion-contradicting resynthesis; listening tests confirmed that its two classes are perceived as intended (4.49 vs. 2.68, $p < 0.001$).
- Designed voice-conversion augmentation of the entrained dyads so that synthetic audio appears in both classes and models must learn affective coordination rather than TTS artifacts.
- Co-developed TRACE, which models how both speakers' emotions evolve over time, using relationship and context as metadata; adding relationship lifts accuracy to **93.47%** vs. 84.50% for DyadFormer.

❖ *[arXiv 2026]* **TurnNat: Automatic Evaluation of Turn-Taking Naturalness in Dyadic Spoken Dialogue** 05/2026-07/2026
- Turn-taking naturalness relies on costly listening tests or per-behavior timing metrics; thus proposed TurnNat, a reference-free metric scoring a whole conversation by its surprisal under a model trained only on natural dialogue.

- Devised turn-boundary units (TBUs) to focus scoring on turn transitions, and built a human-validated benchmark of 1,000 natural–perturbed pairs covering five timing failures.
- Demonstrated that TurnNat picks the natural clip in **88.0%** of pairs, outperforming VAP (80.6%) and unadapted DualTurn (77.5%); code and benchmark are publicly released.

Supervisor: Prof. Philipp Koehn & Prof. Berrak Sisman, Johns Hopkins University

❖ *[ICASSP 2027, under review]* **Holistic Parallel Supervision for Expressive Speech-to-Speech Translation** 03/2026-09/2026

- S2ST pairs rarely match in content, prosody, timing and speaker at once, so existing systems learn expression only indirectly; therefore led the first proof-of-concept study of whether holistic parallel supervision enables better expressive S2ST.
- Built VC-Dub (210k pairs, 434 h), in which professional dubbing supplies human-performed content, prosody and timing, and source-side voice conversion matches the speaker while targets stay real speech.
- Designed DirectS2ST (1.2B), a Moshi-style AR/NAR codec model trained from scratch that jointly predicts text and target speech directly from source speech, conditioned on a source-derived prompt and speaker prefix.
- Demonstrated that VC-Dub improves prosody, speaker similarity and expressiveness for three expressive S2ST systems, most for DirectS2ST (expressiveness MOS **3.42** vs. 2.13 with CVSS-T).

***Research Assistant,* Human Language Technology Center of Excellence (HLTCOE)** 08/2026 - Present

Supervisor: Profs. Kevin Duh, Berrak Sisman, Philipp Koehn, Johns Hopkins University

❖ *[Ongoing]* **Related-Language Adaptation for Low-Resource TTS and Speech Translation** 08/2026 - Present

- Studying whether a related high-resource language can make up for a low-resource language's lack of speech data, by adapting multilingual TTS and training speech translation on the synthetic speech.
- Benchmarked 7 ASR models and adopted the most consistent on both real and synthetic speech as the intelligibility judge.
- Showed that adaptation scales with data: full fine-tuning of VoxCPM2 (2.9B), unseen on the low-resource language, cut CER steadily from 17.70% (zero-shot) to **6.64%** at 85 min, on par with real speech (6.73%) and far below LoRA (14.86%).
- Building on these findings, leading a study on whether neuron-level probing of language representations can guide parameter-efficient adaptation to outperform full fine-tuning under the same data budget.

Independent Study, Johns Hopkins University (with Prof. Philipp Koehn)

❖ *[NAACL 2027, under review]* **KASM: Knowledge-Augmented Sequence Modeling for Document-Level ABSA** 05/2025 - Present

- Proposed KASM to ground document-level aspect sentiment in readable trigger–opinion evidence, which existing models never pinpoint even when it is dispersed, implicit or contradictory across long reviews.
- Built an LLM-extracted knowledge base (86k tuples, >90% human agreement) and a contrastive retriever linking each sentence to an evidence path, whose pseudo-polarities guide FLAN-T5's next-polarity prediction.
- Outperformed THOR, DART, InstructABSA and GPT-5.5 in macro-F1 on BeerAdvocate (**89.80**, +2.73) and TripAdvisor (**87.63**, +1.57), with a 2.70-point gain over DART on long reviews.

***Research Assistant,* Key Laboratory of Machine Intelligence & System Control, Ministry of Education** 07/2023-06/2024

Supervisor: Prof. Teng Li, Shandong University

❖ *[AEI 2025]* **Sequence–Spectrogram Fusion Network for Few-Shot Machinery Diagnosis** 09/2023-06/2024

- Proposed SSFN for data-poor few-shot fault diagnosis, whose sequence and spectrogram branches exchange features at every depth, since single-modality models see either temporal or spectral cues.
- Designed a hybrid-level feature aggregator (global self-attention plus local convolution) that adaptively fuses the complementary modalities, raising accuracy to 97.1% from 89.7% (sequence only).
- Outperformed 10 baselines, e.g. InceptionTime and TimesNet, in fault and impact diagnosis (macro-AUC **0.996**).

❖ *[RICAI 2023, oral]* **Few-Shot Time Series Classification with Time–Frequency Augmentation** 07/2023-09/2023

- Proposed augmenting time–frequency images instead of raw series, by structure-preserving random erasing, since image-style augmentation (cropping, scaling) distorts scarce medical and industrial series.
- Fused each series with its spectrogram in SSNN, reaching **93.75%** F1 on ALS (medical) and **95.48%** on wind-turbine faults.

WORK EXPERIENCE

***Algorithm Engineer Intern,* iFLYTEK CO.LTD.** 07/2024-11/2024

Business Unit: Xunfei Healthcare, Research Institute

❖ ***Xiaoyi* Medical Interrogation LLM Optimization: Data-Centric LLM Fine-Tuning for Patient Persona Slot Extraction**

- To fill missing labels and under-represented slots, designed chain-of-thought prompting with GPT-4o to synthesize dialogues and to screen, correct and standardize pre-annotated data.
- Developed a human-in-the-loop annotation framework coordinating physicians and crowd annotators through automated task routing, cutting rework by **14%** and saving **110** person-hours per month.
- Built a fine-tuning corpus from over 10 GB of dialogues and instruction-tuned *Xiaoyi*, raising average F1 by more than **10%** across 7 patient slots (e.g., medical, allergy and family history, symptoms).

TEACHING EXPERIENCE

- **Course Assistant**, Johns Hopkins University, 2026 Fall, Natural Language Processing (Supervisor: Jason Eisner)
- **Teaching Assistant**, Shandong University, 2024 Spring, Machine Learning (Supervisor: Teng Li)

SKILLS

Programming: Python/MATLAB (advanced), C/C++/SQL/R/Java (intermediate).
Frameworks & Tools: PyTorch, Lightning, Hugging Face Transformers, PyWavelets, SciPy, NLTK, sktime, Matplotlib, WFDB, Git.
Speech: Mimi, w2v-BERT 2.0, Whisper, Demucs, MossFormer2, Sortformer, Seed-VC, Free-VC, human listening tests.

EXTRACURRICULAR HIGHLIGHTS

Student Admission Reviewer for MSE in Computer Science, Department of Computer Science, Johns Hopkins University
WSE Peer Navigator (volunteer), Whiting School of Engineering, Johns Hopkins University
Academic Reviewer Knowledge-based Systems, MethodsX, ICBIR